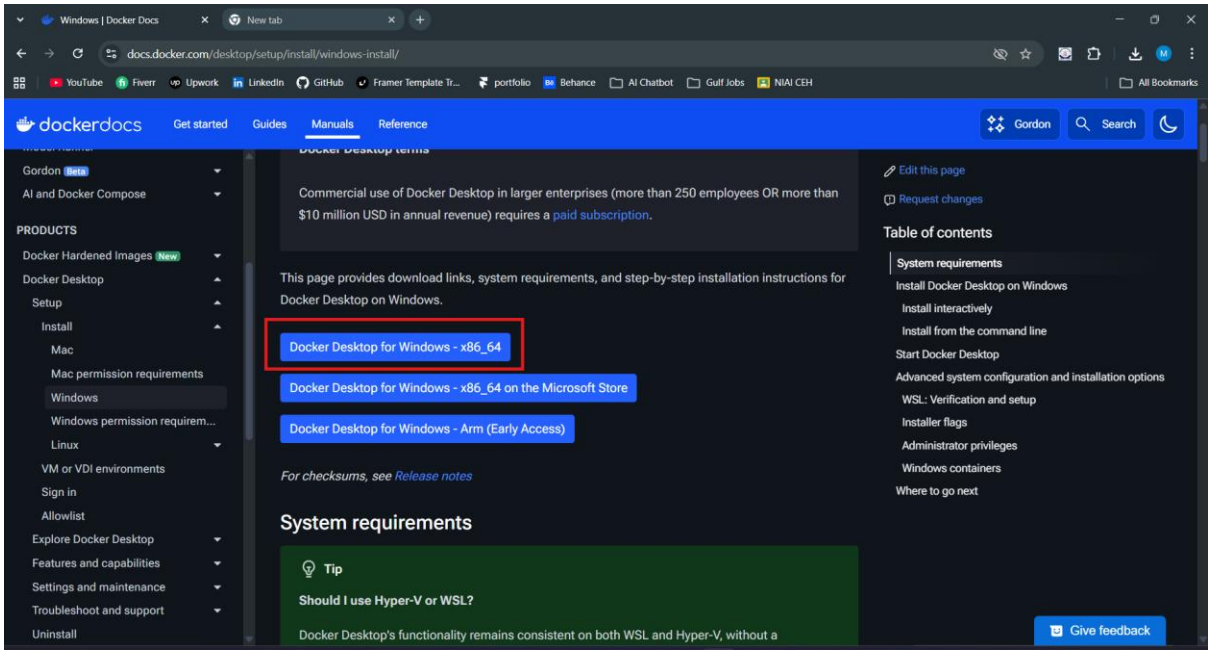
Lab Assignment: Installing Docker and Kali Linux

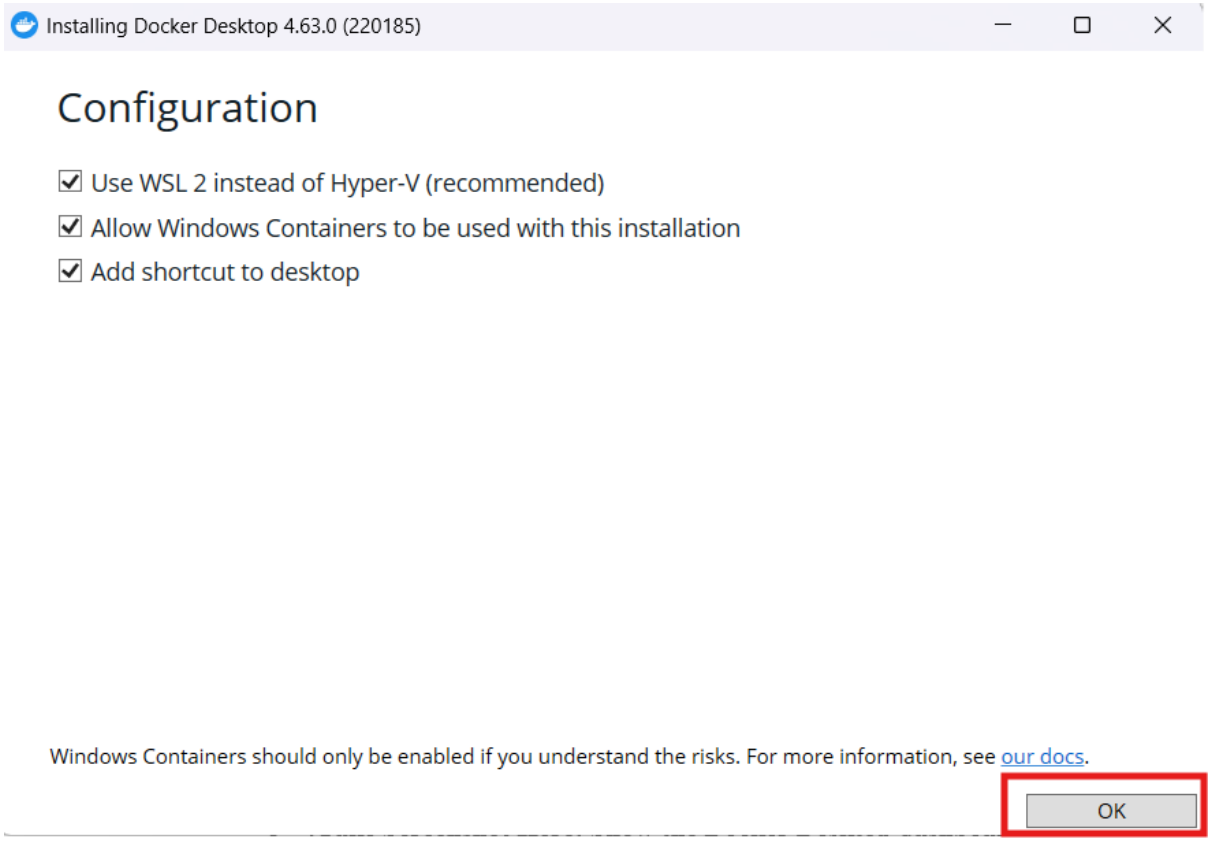
Objective: To successfully install Docker on the host machine and deploy a Kali Linux container using the command-line interface.

Part 1: Installing Docker

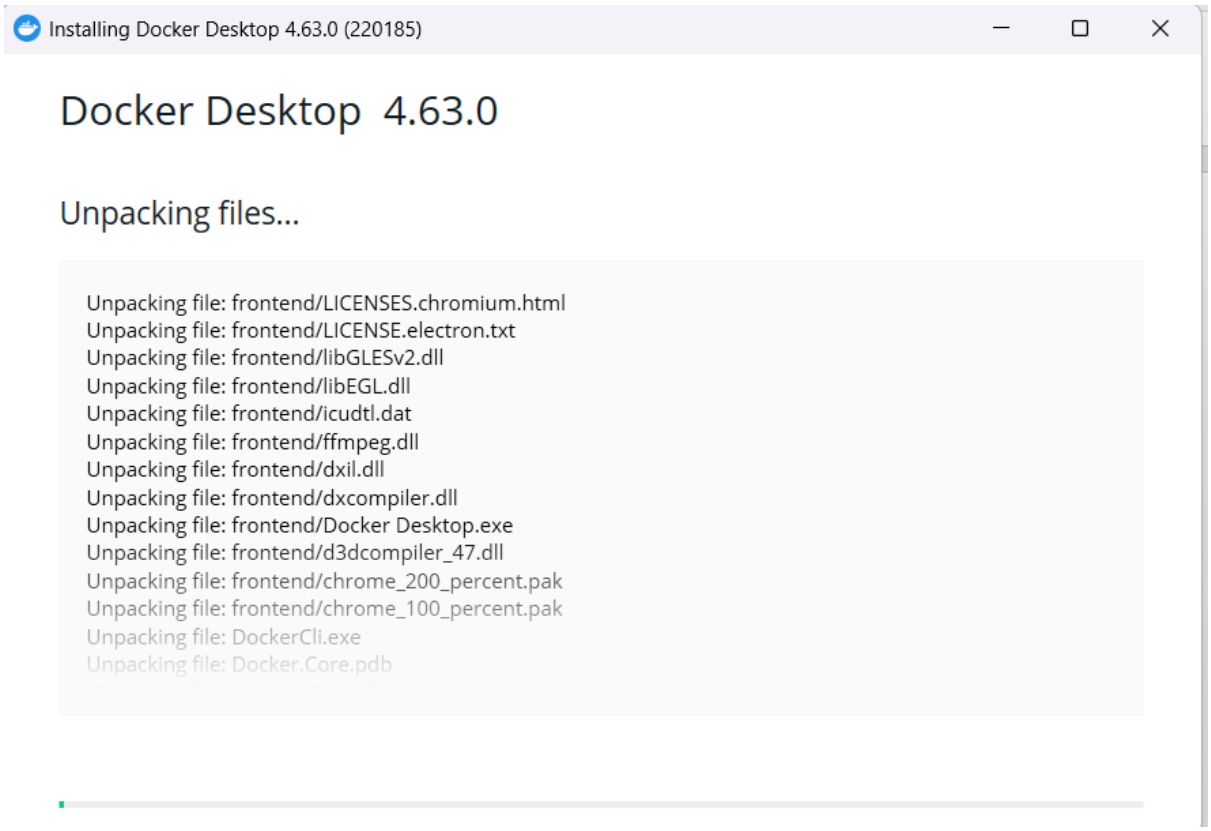
1. **Download Docker Desktop:** Open your web browser and navigate to the official Docker website (docker.com). Download the Docker Desktop installer for your operating system.



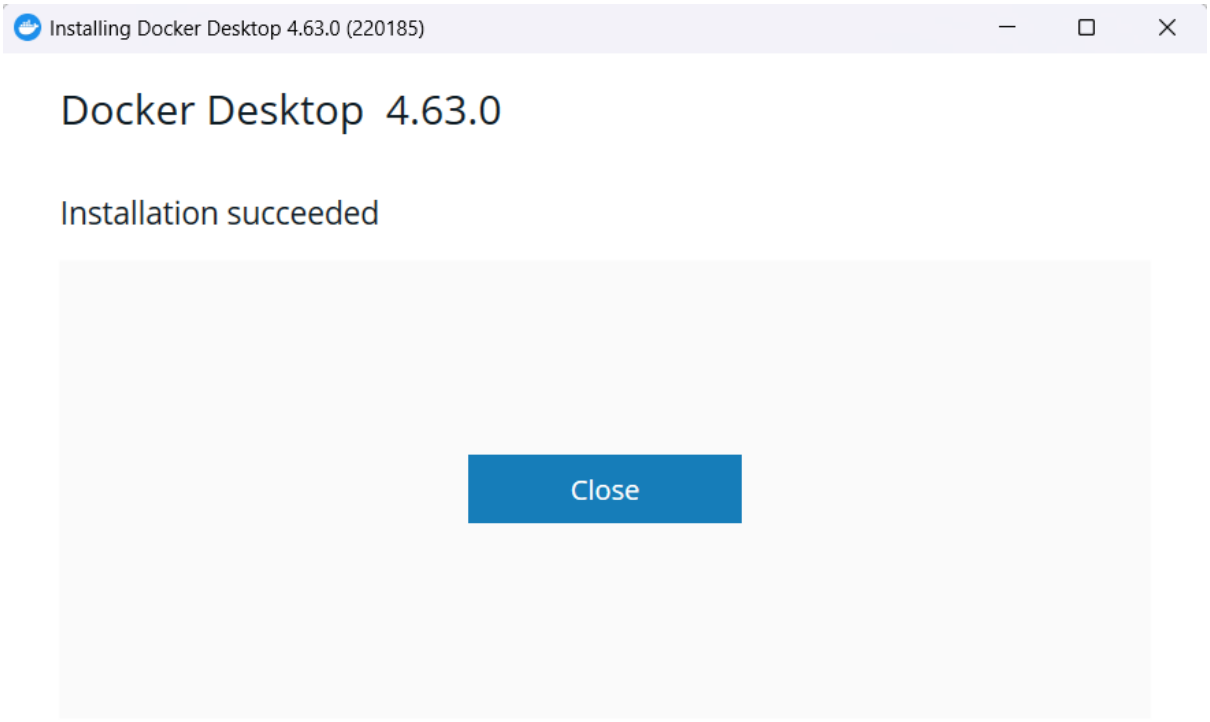
2. **Run the Installer:** Double-click the downloaded setup file to begin the installation process. Keep the default settings checked (especially WSL 2 backend if on Windows) and click "Ok" or "Install".



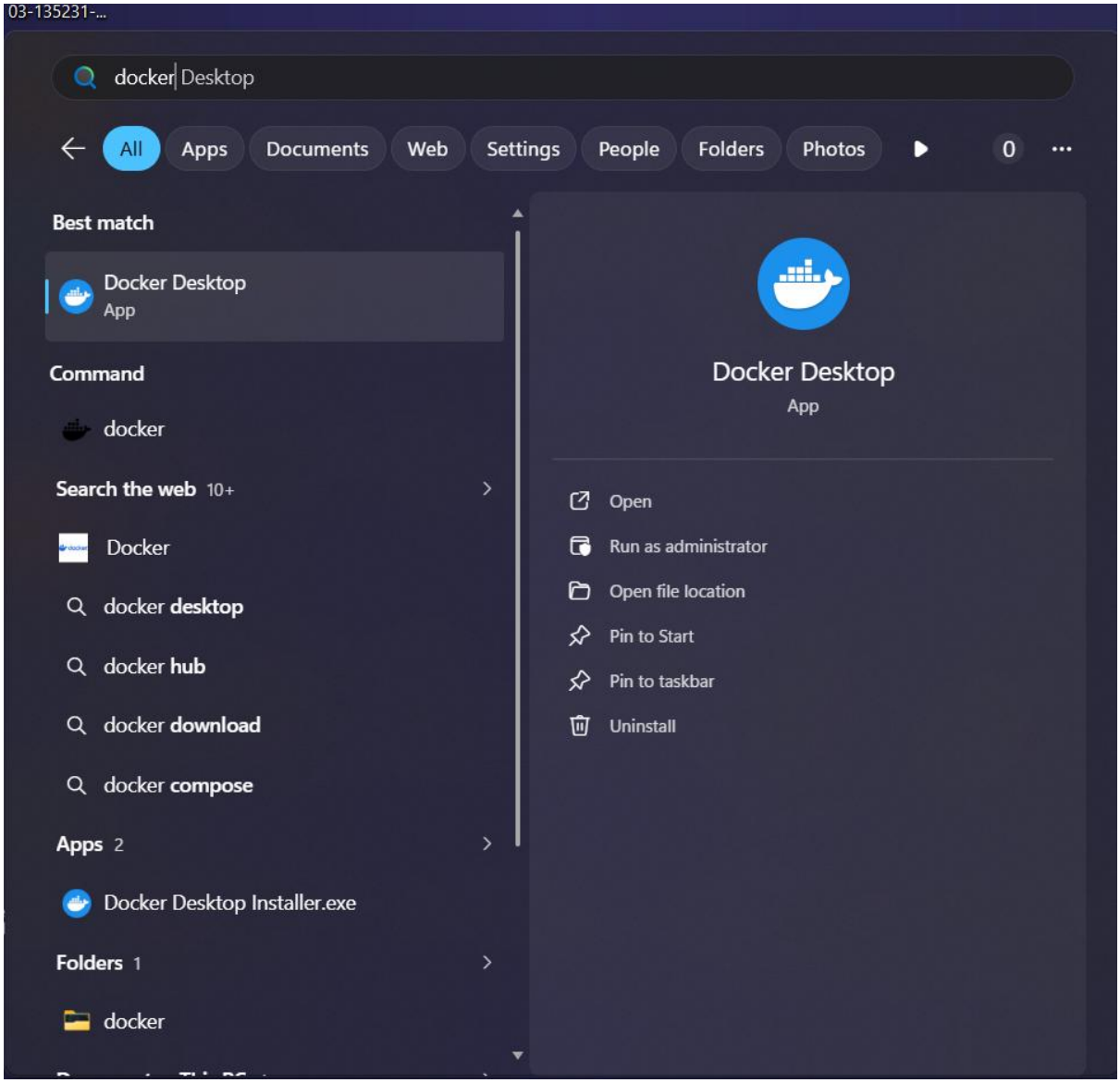
It will take time for completion depending upon internet speed.



3. **Complete Installation:**



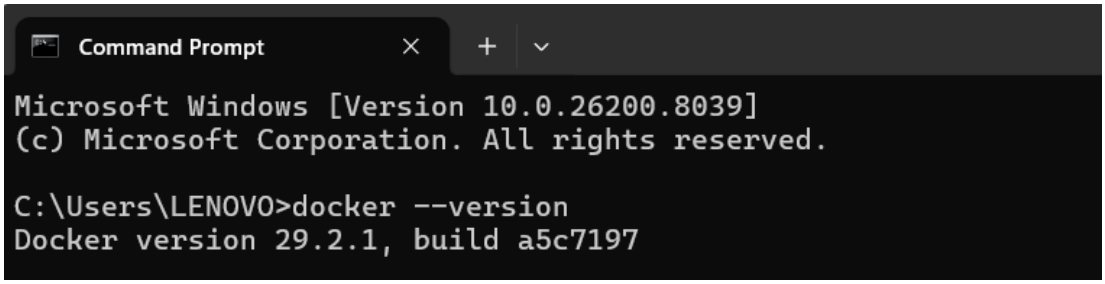
4. **Verify Docker Engine is Running:**



Part 2: Installing and Running Kali Linux via Docker

1. **Open the Command Line:** Open your terminal (Command Prompt or PowerShell on Windows, or Terminal on Linux/Mac).
2. **Verify Docker CLI:** Type the following command to ensure Docker is recognized by your terminal:

```
docker -version
```



3. **Pull the Kali Linux Image:** To download the official Kali Linux image from Docker Hub, enter the following command:

```
docker pull kalilinux/kali-rolling
```

```
C:\Users\LENOVO>docker pull kalilinux/kali-rolling
Using default tag: latest
latest: Pulling from kalilinux/kali-rolling
194d70789344: Downloading [====>                                          ]  4.194MB/52.22MB
```

4. **Run the Kali Linux Container:** Once the download is complete, start an interactive container using the downloaded image by typing:

```
docker run -it kalilinux/kali-rolling /bin/bash
```

```
C:\Users\LENOVO>docker run -it kalilinux/kali-rolling /bin/bash
┌──(root㉿20ba4d4836a7)-[/]
└─#
```

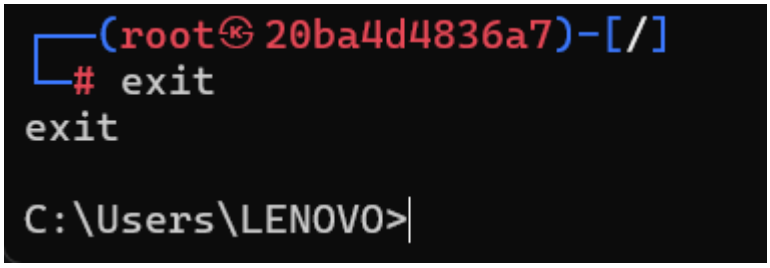
Explanation: The `-it` flags allow you to interact with the container through a terminal, and `/bin/bash` opens the Bash shell inside Kali.

Part 3: Verifying the Kali Linux Installation

1. **Check OS Information:** To prove that you are currently inside the Kali Linux environment, run the following command to display the OS release data:

```
cat /etc/os-release
```

```
┌──(root㉿20ba4d4836a7)-[/]
└─# cat /etc/os-release
PRETTY_NAME="Kali GNU/Linux Rolling"
NAME="Kali GNU/Linux"
VERSION_ID="2026.1"
VERSION="2026.1"
VERSION_CODENAME=kali-rolling
ID=kali
ID_LIKE=debian
HOME_URL="https://www.kali.org/"
SUPPORT_URL="https://forums.kali.org/"
BUG_REPORT_URL="https://bugs.kali.org/"
ANSI_COLOR="1;31"
```

2. **Exit the Container:** When you are finished with your lab, you can exit the Kali container by simply typing:

```
exit
```